Longitudinal Data Analysis
Assignment 1: Continuous Longitudinal Data

Luis César Herrera Paltán, Daniel Sanson,
Maryam Shojaei Shahrokhabadi, Nele Verhulst

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1 Introduction

The most common cause of dementia in Belgium is Alzheimer's disease. This study followed 1253 patients starting from their first diagnosis of Alzheimer's disease. Baseline patient data included age (AGE), sex (SEX), education level (EDU), body mass index (BMI), whether the person had a paid job (JOB), activities of daily living (ADL), and whether they live in a nursing home (WZC). Patients were followed up through 25 clinical centers for baseline measurements and then annually over six years. Four variables were measured at each of the seven time points: two cognitive scales (CDR-SB and BPRS) and two possible biomarkers (ABPET and TAUPET). The main research question was how Brief Psychiatric Rating Scale (BPRS) changed over time, and how it related to the patient characteristics at baseline and the baseline measurements of CDR-SB, ABPET, and TAUPET. Specific research questions were the following:

1. What are the implications of the graphical exploration of mean, variance, and correlation structure for the statistical modeling?
2. Are the conclusions of the analysis of the summary statistics for BPRS consistent with the conclusions of a linear mixed model?
3. Are the conclusions of a multivariate model consistent with the conclusions of a linear mixed model?
4. Can an explicit two-stage analysis give an initial impression of trends in BPRS and the effect of covariates on those trends?
5. How do the subject-specific intercepts/slopes from a random-effects model compare to those from the multivariate model?

2 Statistical methods

2.1 Data preparation

Positron Emission Tomography (PET) that targets amyloid-plaques and tau proteins is used in the diagnosis of Alzheimer's disease (Wang et al., 2024). Both are included in the dataset as the standardized uptake value ratio of a specific radioactive tracer. For purpose of the analysis, the Amyloid-Positron Emission Tomography (ABPET) and Tau Positron Emission Tomography (TAUPET) at baseline were categorized. For ABPET, two groups were used with values (1) < 2.5, (2) ≥ 2.5. The TAUPET values were coded as (1) 1.9 and (2) > 1.9.

The Clinical Dementia Rating Sum of Boxes (CDRSB) is used for dementia staging, with four main staging categories proposed by O'Bryant et al. (2008): (1) $0.5 - 4.0$ questionable cognitive impairment, (2) $4.5 - 9.0$ mild dementia, (3) $9.5 - 15.5$ moderate dementia, ≥ 16.0 severe dementia. For consistency with assignment 2, the groups defined for this report were (1) ≤ 10 and (2) > 10. Age groups were defined as (1) ≤ 67, (2) $68 - 77$, and (3) ≥ 78. For BMI, common categories of normal ($18.5 \leq BMI \leq 24.9$), overweight ($25 \leq BMI \leq 29.9$) and obese ($BMI \geq 30$) were used.

2.2 Exploration of data, mean, variance and correlation structure

The exploratory analysis included all patients, so also the ones with only baseline data. Individual profiles of BPRS score over time were revised for 50 randomly selected patients to get a first feel for the evolution of the main response variable. The distribution of patients over different groups was revised for all covariates, and the retention rate calculated. The mean structure of BPRS over years was graphed overall, as well as for the patients grouped for the different covariates. The variance structure of BPRS was revised by visualizing the standard deviation of the mean and smoothing the variance function over the squared residuals. Because the variance looked approximately constant, the semivariogram was plotted to study the serial correlation function. The correlation structure was examined further by looking at the scatter plot matrix and pairwise correlations of BPRS time points.

2.3 Analysis through the use of summary statistics

To compare the evolution of BPRS over time among subjects, the vector of repeated measurements for each individual was summarized separately using three summary statistics: the area under the curve (AUC), endpoint values, and increments. To ensure comparability across subjects, these summary statistics were calculated only for subjects with complete data (511 patients), excluding those with missing values.

For the AUC analysis, the area under the response curve was calculated for each subject, denoted as AUC_i. In the increments analysis, subject-specific changes were calculated as the difference between the last and first measurements, $\Delta f_i = y_{in} - y_{i1}$.

Subsequently, the summary statistics AUC_i, Δy_i and the last measurements of each subject (y_{ni}) were analyzed using an analysis of covariance (ANCOVA) to compare individual evolution (using their summary statistics) while accounting for covariates potentially affecting BPRS, as identified during the exploratory phase (i.e. sex, age group, nursing home status, baseline CDRSB, and baseline ABPET). For this and all subsequent analyses, the significance level was set to $\alpha = 0.05$. The ANCOVA model can be expressed as:

$$\text{Summary Statistic}_i = \beta_0 + \beta_1 \,\text{Sex}_i + \beta_2 \,\text{WZC}_i + \beta_3 \,\text{AgeGroup2}_i + \beta_4 \,\text{AgeGroup3}_i + \beta_5 \,\text{CDRSB}_i + \beta_6 \,\text{ABPET}_i + \epsilon_i$$

$$i = 1, \ldots, m \quad \text{(patients with complete measurements)}$$

where:

- β: Regression parameters
- Sex: Dummy variable with Male as reference group (0: Male, 1: Female).
- WZC: Dummy variable with non-nursing home residents as reference (0: non-residents, 1: nursing home residents).
- AgeGroup2: Dummy variable with AgeGroup1 (≤ 67) as reference (0: Age ≤ 67, 1: age 68-77).
- AgeGroup3: Dummy variable with AgeGroup1 (≤ 67) as reference (0: Age ≤ 67, 1: age ≥ 78).
- CDRSB: Dummy variable with group with CDRSB ≤ 10 as reference (0: CDRSB ≤ 10, 1: CDRSB > 10).
- ABPET: Dummy variable with group with ABPET < 2.5 as reference (0: ABPET < 2.5, 1: ABPET ≥ 2.5).

2.4 Analysis with a multivariate model

Only patients with at least two measurements (1106 patients) were included in the analysis with the multivariate model, the two-stage analysis, and the random-effects model to allow comparison of the results of these different models. A general multivariate model was fit for $\mathbf{Y}_i$, the vector of 7 repeated measurements of BPRS for the i-th patient:

$$\mathbf{Y}_i = X_i \boldsymbol{\beta} + \boldsymbol{\epsilon}_i \tag{1}$$

with

- X_i the matrix of covariates including the time point, age group, sex, nursing home, baseline ABPET and baseline CDRSB dummy variables defined as above for the ANCOVA model
- $\boldsymbol{\beta}$ the vector of regression parameters
- $\boldsymbol{\epsilon}_i$ the vector of error components with $\boldsymbol{\epsilon}_i \sim \mathcal{N}(\mathbf{0}, \Sigma)$

First, the model was fit with additive main effects for age group, sex, nursing home, baseline ABPET and baseline CDRSB and the interaction of each of those with year. The most parsimonious mean structure was identified through likelihood ratio tests for nested models, first reducing the covariates with a slope with year, and then checking for the covariates that were removed from the slopes, which main effects could be omitted from the model. Finally, two alternatives for the covariance structure were compared to the unstructured covariance matrix: heterogeneous Toeplitz, and heterogeneous first-order autoregressive.

2.5 Two-stage analysis

A two-stage modeling approach was used to analyze subject-specific longitudinal profiles, which can be approximated using linear regression model. The model formulation had two stages. In the first stage, a linear regression model was fitted for each subject to capture individual trajectories over time. To choose between the reduced (BPRS $\sim$ year) or a full model (BRPS $\sim$ year + $year^2$), the F-meta test for model extension was used. This test aggregates, across all subjects, the individual F-tests that compare the reduced model to the full model. Additionally, R^2_{Meta} was computed, which evaluates the proportion of total variability explained across subjects. The second stage of the analysis modeled subject-specific intercepts and time effects (slopes) as a function of covariates.

Stage 1:

Let Y_{ij} denote the BPRS measurement taken for subject i ($i = 1, \ldots, n$) at year j ($j = 1, \ldots, n_i$) .

$$Y_{ij} = \beta_{1i} + \beta_{2i}\, year_{ij} + \varepsilon_{ij}; \qquad j = 1, \ldots, n_i, \tag{2}$$

where β_{1i} and β_{2i} are the subject-specific intercept and slope, respectively, and the within-subject errors are assumed to

$$\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2).$$

Equivalently, in vector form, for $\mathbf{Y}_i$, the vector of 7 repeated measurements of BPRS for the i-th patient:

$$\mathbf{Y}_i = \mathbf{Z}_i \boldsymbol{\beta}_i + \boldsymbol{\varepsilon}_i,$$

where $\boldsymbol{\beta}_i = (\beta_{1i}, \beta_{2i})^\top$, and $\mathbf{Z}_i$ is the $n_i \times 2$ design matrix with rows $(1,\, t_{ij})$.

This stage provides individual estimates of regression parameters that describe each subject's longitudinal profile, capturing the within-subject variability in repeated BPRS measurements. These estimates are then used in Stage 2 to model the variability between subjects.

Stage2

Between-subject variability was then studied by relating the betas to known covariates (i.e. sex, age group, nursing home status, baseline CDRSB, and baseline ABPET).

$$\beta_{1i} = \beta_1 + \beta_2 \text{Female}_i + \beta_3 \text{WZC}_i + \beta_4 \text{AgeGroup2}_i + \beta_5 \text{AgeGroup3}_i + \beta_6 \text{CDRSB1}_i + \beta_7 \text{ABPET1}_i + b_{1i},$$
$$\beta_{2i} = \beta_8 + \beta_9 \text{Female}_i + \beta_{10} \text{WZC}_i + \beta_{11} \text{AgeGroup2}_i + \beta_{12} \text{AgeGroup3}_i + \beta_{13} \text{CDRSB1}_i + \beta_{14} \text{ABPET1}_i + b_{2i}.$$

Where:

1. All variables are defined as in Section 2.3.
2. $\beta_1, \ldots, \beta_7$ determine the average intercept for each group, after correcting for these variables.
3. $\beta_8, \ldots, \beta_{14}$ are the average linear time effects for each group, after correcting for these variables.
4. b_{1i} and b_{2i} are subject-specific random effects, assumed to follow

$$\mathbf{b}_i = (b_{1i}, b_{2i})^\top \sim \mathcal{N}(\mathbf{0}, \mathbf{D}).$$

2.6 Analysis with a random-effects model

A Linear Mixed-Effects (LME) model was used to account for the relationships in repeated measurements from the same subjects. The best model was selected using a step-by-step approach. First, a comprehensive preliminary mean structure ($\mathbf{X}_i\boldsymbol{\beta}$) was specified, including covariates such as sex, age group, nursing home status, baseline CDRSB, and ABPET. Second, a preliminary random-effects structure ($\mathbf{Z}_i\mathbf{b}_i$) was incorporated, allowing for a random intercept, a random slope, and a quadratic time effect. Third, the residual covariance structure ($\boldsymbol{\epsilon}_i$) was evaluated using Restricted Maximum Likelihood (REML). Based on the observed decrease in correlations over time, Unstructured, ARH(1), and Toeplitz structures were compared, with the Unstructured covariance ultimately retained.The full model is formulated as:

$$\begin{aligned}\text{BPRS}_{ij} = {} & \beta_0 + \beta_1\text{Sex}_i + \beta_2\text{AgeGroup2}_i + \beta_3\text{AgeGroup3}_i + \beta_4\text{WZC}_i + \beta_5\text{CDRSB}_i + \beta_6\text{ABPET}_i \\ & + (\beta_7\text{Sex}_i + \beta_8\text{AgeGroup2}_i + +\beta_9\text{AgeGroup3}_i + \beta_{10}\text{WZC}_i + \beta_{11}\text{CDRSB}_i + \beta_{12}\text{ABPET}_i)\cdot\text{Year}_{ij} \\ & + (\beta_{13}\text{Sex}_i + \beta_{14}\text{AgeGroup2}_i + \beta_{15}\text{AgeGroup3}_i + \beta_{16}\text{WZC}_i + \beta_{17}\text{CDRSB}_i + \beta_{18}\text{ABPET}_i)\cdot\text{Year}_{ij}^2 \\ & + b_{0i} + b_{1i}\text{Year}_{ij} + b_{2i}\text{Year}_{ij}^2 + \epsilon_{(1)i} + \epsilon_{(2)i},\end{aligned} \tag{3}$$

where:

- b_{0i}, b_{1i}, b_{2i} are the subject-specific random intercept, slope, and quadratic slope, assumed to follow

$$\mathbf{b}_i = (b_{0i}, b_{1i}, b_{2i})^\top \sim N(\mathbf{0}, \mathbf{D}).$$

- $\epsilon_{(1)i}, \epsilon_{(2)i}$ are error components and serial correlation respectively.
- The β coefficients represent population-average fixed effects for the corresponding covariates and their interactions with linear and quadratic time.
- All variables are defined as in Section 2.3.

Finally, reductions of both the random-effects and mean structures were performed sequentially, removing non-significant components to obtain a parsimonious final model. Subject-specific intercepts and slopes were calculated as the best linear unbiased predictors (BLUPs) from the LME model and compared to estimates from the two-stage analysis.

3 Results and discussion

3.1 Exploration of data, mean, variance and correlation structure

Individual profiles of BPRS by year for 50 randomly selected patients indicated linear increases and mostly parallel lines (Figure 1, Left). This suggested that a linear model was appropriate to describe the evolution of BPRS score per patient and indicated that the variability between patients was greater than the variability within patients. On average, the BPRS score was 75.4 at baseline and increased to 107.0 in year 6 (Figure 1, right).

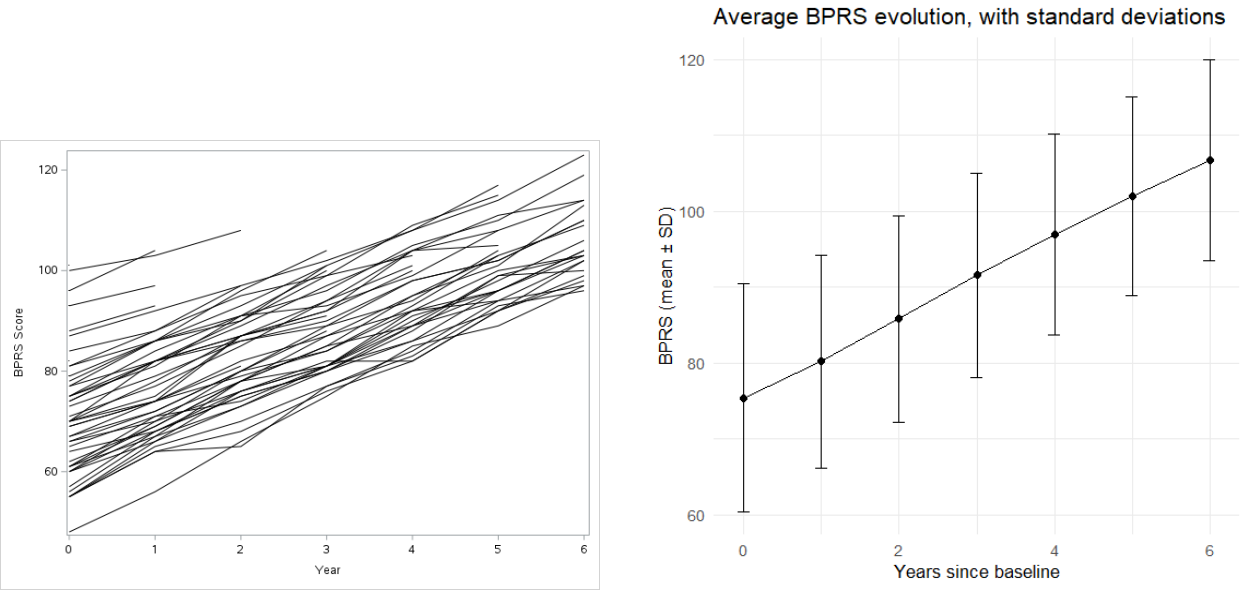


Figure 1: BPRS scores over years (Left) Individual profiles for 50 randomly selected patients. (Right) Average BPRS scores, with error bars indicating standard deviation.

In total, 40.8% of the participants remained in the study in year 6, indicating a high overall dropout rate and that trends at later times should be interpreted with caution. The patient group had approximately the same number of women (637 patients) and men (616 patients), but the retention rate was about a quarter for women (26.7%) while more than half of the male patients stayed in the study until year 6 (55.4%; Table 1). The mean structure was similar for male and female patients, with only a slightly lower mean BPRS score for male patients from baseline to year 4 (not shown).

Table 1: Retention over time by subgroups

Variable	Group	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5	Year 6
Sex	Male	616 (100.0%)	583 (94.6%)	551 (89.4%)	506 (82.1%)	463 (75.2%)	409 (66.4%)	341 (55.4%)
	Female	637 (100.0%)	523 (82.1%)	463 (72.7%)	406 (63.7%)	342 (53.7%)	241 (40.0%)	170 (26.7%)
WZC	No nursing	766 (100.0%)	739 (96.5%)	721 (94.1%)	686 (89.6%)	626 (81.7%)	559 (73.0%)	465 (60.7%)
	Nursing home	487 (100.0%)	367 (75.4%)	293 (60.2%)	221 (45.4%)	151 (31.0%)	93 (19.1%)	46 (9.5%)
Age group	≤67 years	376 (100.0%)	375 (99.7%)	375 (99.7%)	370 (98.4%)	362 (96.3%)	347 (92.3%)	316 (84.0%)
	68–77 years	576 (100.0%)	549 (95.3%)	514 (89.2%)	456 (79.2%)	374 (64.9%)	288 (50.0%)	186 (32.3%)
	≥78 years	301 (100.0%)	182 (60.5%)	125 (41.5%)	81 (26.9%)	41 (13.6%)	17 (5.7%)	9 (3.0%)
BMI group	Normal (18.5–24.9)	451 (100.0%)	405 (89.8%)	379 (84.0%)	346 (76.7%)	304 (67.4%)	262 (58.1%)	204 (45.2%)
	Overweight (25–29.9)	759 (100.0%)	666 (87.7%)	604 (79.6%)	534 (70.4%)	451 (59.4%)	372 (49.0%)	295 (38.9%)
	Obese (≥30)	43 (100.0%)	35 (81.4%)	31 (72.1%)	27 (62.8%)	22 (51.2%)	18 (41.9%)	12 (27.9%)
Job	No job	1149 (100.0%)	1002 (87.2%)	910 (79.2%)	803 (69.9%)	673 (58.6%)	548 (47.7%)	407 (35.4%)
	Job	104 (100.0%)	104 (100.0%)	104 (100.0%)	104 (100.0%)	104 (100.0%)	104 (100.0%)	104 (100.0%)
Education	Primary	156 (100.0%)	139 (89.1%)	128 (82.1%)	120 (76.9%)	104 (66.7%)	91 (58.3%)	59 (37.8%)
	Lower secondary	246 (100.0%)	217 (88.2%)	204 (82.9%)	174 (70.7%)	148 (60.2%)	121 (49.2%)	98 (39.8%)
	Upper secondary	385 (100.0%)	331 (86.0%)	299 (77.7%)	275 (71.4%)	228 (59.2%)	194 (50.4%)	153 (39.7%)
	Higher education	466 (100.0%)	419 (89.9%)	383 (82.2%)	345 (74.0%)	299 (64.2%)	256 (54.9%)	201 (43.1%)
TAUPET	1.9	1197 (100.0%)	1089 (91.0%)	1008 (84.2%)	907 (75.8%)	777 (64.9%)	652 (54.5%)	511 (42.7%)
	> 1.9	56 (100.0%)	17 (30.4%)	6 (10.7%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
ABPET	< 2.5	845 (100.0%)	820 (97.0%)	790 (93.5%)	734 (86.9%)	668 (79.1%)	590 (69.8%)	483 (57.2%)
	≥ 2.5	408 (100.0%)	286 (70.1%)	224 (54.9%)	173 (42.4%)	109 (26.7%)	62 (15.2%)	28 (6.9%)
CDRSB	≤ 10	889 (100.0%)	797 (89.7%)	730 (82.1%)	652 (73.3%)	563 (63.3%)	461 (51.9%)	351 (39.5%)
	> 10	364 (100.0%)	309 (84.9%)	284 (78.0%)	255 (70.1%)	214 (58.8%)	191 (52.5%)	160 (44.0%)
Overall		**1253 (100%)**	**1106 (88.3%)**	**1014 (81.0%)**	**907 (72.4%)**	**777 (62.0%)**	**652 (52.0%)**	**511 (40.8%)**

Almost 40% of the patients were living in a nursing home at the beginning of the study, and retention rate in that group was only 9.5%, while for other patients it was 60.7%. Nursing home patients scored about 15 points higher BPRS on average at each time point, compared to the patients not living in a nursing home (Figure 2).

About 30% of patients were at or below retirement age (67 years old) at the start of the study, and about one quarter was 78 or older (Table 1). As expected, the retention rate was the lowest in the oldest group, with only 9 patients still in the study in year 6. A scatter plot of BPRS versus baseline age showed that BPRS increased with baseline age and illustrated that dropout was higher for older patients (Figure 3a). The mean structure of BPRS over years showed parallel lines for the three age groups, with BPRS consistently higher for older patient groups (Figure 3b).

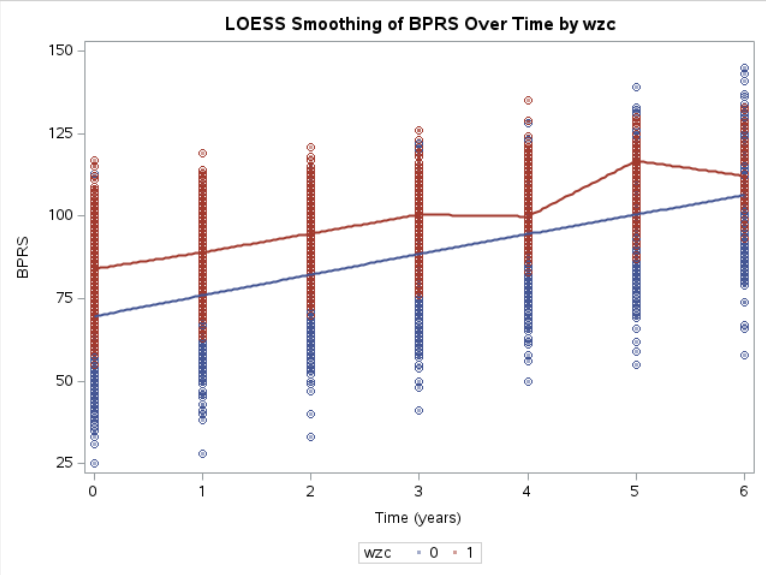
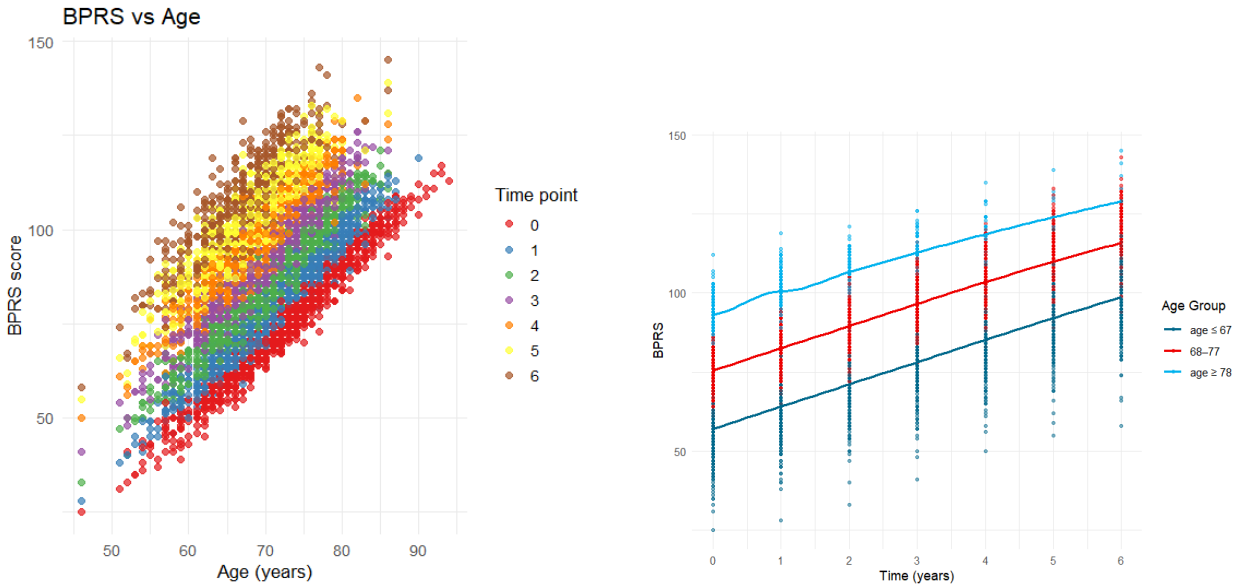


Figure 2: BPRS evolution over years for patients not living in nursing homes (blue) and living in nursing homes (red).



(a) BPRS vs age

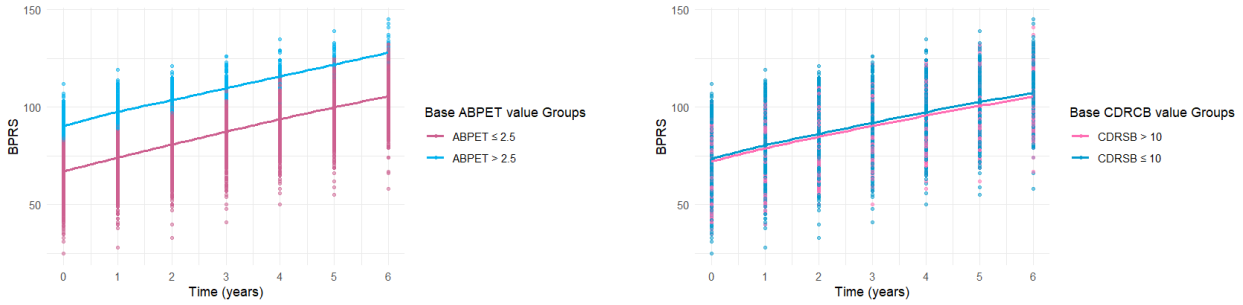
(b) BPRS by age group

Figure 3: Visualizations of BPRS and patients' age. (Left) BPRS scores versus baseline age; (Right) BPRS evolution by age groups.

According to BMI, 36.0% of patients had normal weight, 60.6% were overweight, and the remaining 3.4% obese. The mean structure of BPRS over years was similar in the three BMI groups, with exception of slightly lower BPRS in the obese group in year 5 and 6, when the group became very small (not shown). Only 104 patients had a job at the beginning of the study, making up 8.3% of patients. Retention rate and the mean structure of BPRS versus year were similar for the four education level groups (Table 1; not shown).

For TAUPET at baseline, 1197 patients had a value of 1.9. The 56 patients with a TAUPET value > 1.9 all dropped out of the study by year 3 (Table 1), so it was not possible to assess how the baseline TAUPET

value related to the BPRS change over time. For ABPET at baseline, two thirds of patients had low values, with the majority concentrated at ABPET value of 2.0. Of the one third of patients with values > 2.5, most had an ABPET of 3.0. The mean BPRS was the lowest in the group with baseline ABPET < 2.5 at each time point (Figure 4a). The BPRS mean structure was similar in the two baseline CDRSB groups (Figure 4b).

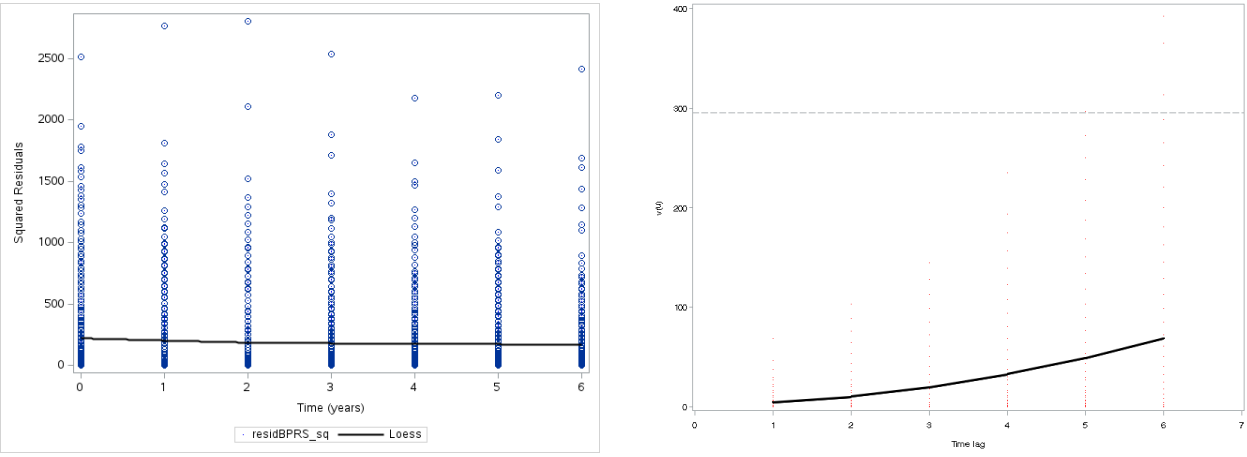


(a) Two groups of ABPET at baseline

(b) Two groups of CDRSB at baseline

Figure 4: BPRS evolution over years for patients with different baseline characteristics.

The average evolution of BPRS over years showed a linear increase with approximately constant variance (Figure 1, right). The variance function fitted to the squared residuals showed a slight decrease, which could be related to the high drop-out rate (Figure 5, left). This constant variance suggested a model without random slopes would be appropriate. Because of the assumed constant variance, the semi-variogram was used to explore the correlation structure of repeated BPRS measurements. The increase in the semi-variogram with time lag suggested the presence of serial correlation (Figure 5b). The scatter plot matrix with pairwise correlations of BPRS showed that the scatter plots were similar within diagonal bands, implying that time points with the same time difference were similarly correlated (e.g. the correlation between measurements in year 2 and 4 was similar to the correlation between year 3 and 5) (Figure 6). The correlations also seemed different between diagonal bands. Therefore, the heterogeneous Toeplitz model and heterogeneous AR(1) model could be tried as alternatives to an unstructured covariance matrix.



(a) Smoothed variance function

(b) Semi-variogram

Figure 5: Graphs examining the variance structure of BPRS.

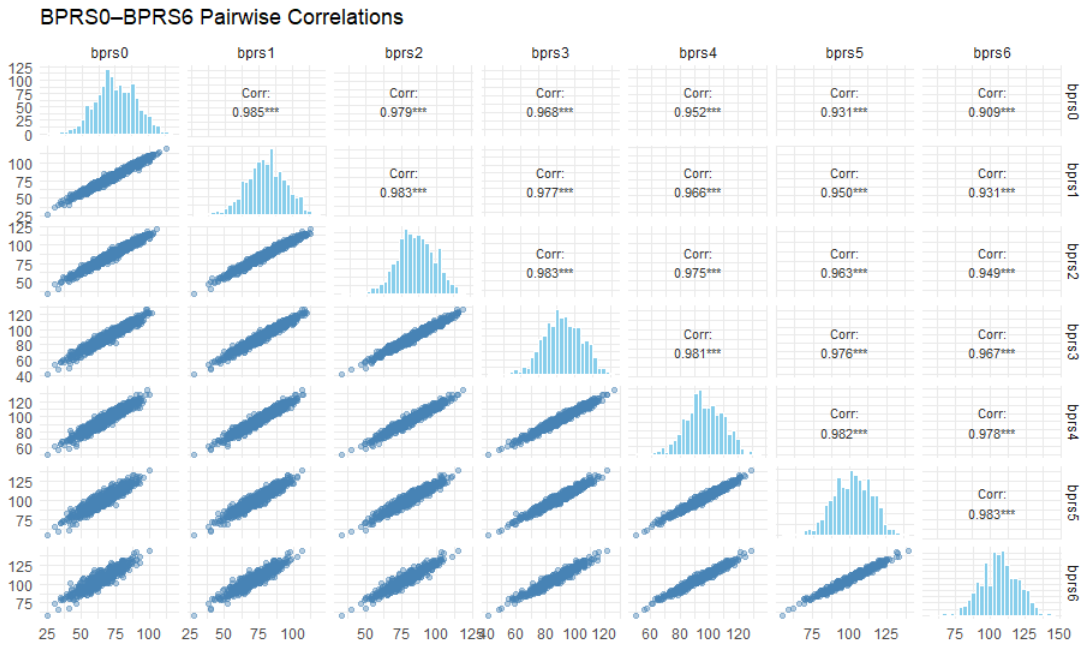


Figure 6: Scatter plot matrix and correlation coefficients for BPRS scores.

3.2 Analysis through the use of summary statistics

Analyses of summary statistics rather than the data of all seven time points allow for using a general linear model, but result in a loss of information. Since BPRS increased linearly with year, patients with partial data (e.g. data that end in year 4) had a lower AUC than patients with complete data. Therefore, only patients with complete data (40.8% of patients) were included in the summary statistics analysis. For the included patients, only one of the seven BPRS measurements was used for the endpoint analysis, and only two of the seven in the increment analysis, marking an even larger loss of information.

All three methods (AUC, endpoint values, and increments) indicated that sex did not affect changes in BPRS over time (Table 2). However, for the the other covariates, results were mixed.

For age group, different summary-statistics approaches produced noticeably different estimates and p-values. For example, the age group 68–77 showed a positive and significant effect in the AUC analysis (Estimate = 99.40(SE = 4.44), $p < .0001$) and in the endpoint analysis (Estimate = 16.92 (SE = 0.87), $p < .0001$), whereas the increment analysis—which directly measures change from baseline to year 6—yielded a small, nonsignificant effect (Estimate = 0.43 (SE = 0.52), p = 0.40). The AUC captures the cumulative level of BPRS across all seven years; therefore, the large AUC estimate suggested that patients aged 68–77 consistently had higher BPRS values throughout the study period. In contrast, the increments reflect change from baseline to year 6 and were similar between age groups, indicating that although older patients start higher (and therefore accumulate a larger AUC), their rate of change over time was similar to that of younger patients. The endpoint uses only one measurement and does not account for the trajectory over time, so it could not adequately characterize change or progression. Thus, while endpoints detect cross-sectional differences at year 6, they were not valid for assessing longitudinal change in this study.

For the group with baseline CDRSB higher than 10, the AUC and the analysis of increments both showed a higher increase in BPRS compared to the low CDRSB group, for patients with the same sex, age group nursing home status and baseline ABPET, but the analysis of endpoints had a negative estimate (Table 2). For the baseline ABPET, the AUC and endpoints were higher than in the low ABPET group, but the increments were unaffected.

Table 2: Summary of results from the analysis of summary statistics

Parameter	**Analysis of AUC**		**Analysis of Increments**		**Analysis of Endpoints**	
	Estimate (SE)	p-value	Estimate (SE)	p-value	Estimate (SE)	p-value
Intercept	461.62 (4.34)	< 0.0001	40.76 (0.51)	< 0.0001	100.37 (0.64)	< 0.0001
Sex (Female)	-2.58 (4.36)	0.55	-0.28 (0.51)	0.58	-0.73 (0.86)	0.40
Age 68–77	99.40 (4.44)	< 0.0001	0.43 (0.52)	0.40	16.92 (0.87)	< 0.0001
Age ≥ 78	119.54 (19.28)	< 0.0001	-0.89 (2.27)	0.70	19.71 (3.80)	< 0.0001
Nursing home	33.80 (7.22)	< 0.0001	0.84 (0.84)	0.32	5.93 (1.42)	< 0.0001
CDRSB > 10	-12.01 (4.43)	0.007	-2.11 (0.52)	< 0.0001	-3.09 (0.87)	< 0.0001
ABPET ≥ 2.5	62.11 (11.29)	< 0.0001	-0.58 (1.33)	0.66	9.83 (2.22)	< 0.0001

3.3 Analysis with a multivariate model

The first model estimated 51 parameters: 28 covariance parameters, 1 intercept, 11 for the main effects of age (3), sex (2), nursing home (2), baseline ABPET (2) and CDRSB (2), and 11 slopes for the same covariates with year. Models 2 to 6 were tested to find the most parsimonious structure for the slopes, resulting in a model that only included the interaction of year and CDRSB (Table 3). For the main effects, only sex was not needed in the model (p=0.86). Because the exploratory analysis pointed to similar correlations between points with the same time difference, two alternatives were tested to reduce the covariance structure, but both resulted in a model that fit the data less well than the unstructured covariance matrix. Therefore, the most parsimonious model that could be identified had an unstructured covariance matrix, main effects of age, nursing home, baseline ABPET and CDRSB, and different slopes with year for the two CDRSB groups.

Table 3: Overview of likelihood ratio tests for model reduction of the multivariate model. Models 2–8 test reductions in the mean structure; Models 9–10 test reductions in the covariance structure.

M. No.	Model description (Mean structure; covariance structure)	Par.	-2ℓ	Ref.	G^2	df	P-value
1	Main effects and interaction with year for Age, Sex, WZC, ABPET, CDRSB; UN	51	29515.2				
2	Model 1, removing CDRSB×year; UN	49	29534.7	1	19.5	2	< 0.0001
3	Model 1, removing ABPET×year; UN	49	29515.2	1	0	2	1
4	Model 3, removing WZC×year; UN	47	29519.3	3	4.1	2	0.1287
5	Model 4, removing Age×year; UN	44	29520.0	4	0.7	3	0.8732
6	Model 5, removing Sex×year; UN	42	29520.1	5	0.1	2	0.9512
7	Model 6, removing WZC; UN	40	29737.3	6	62.1	2	< 0.0001
8	Model 5, removing Sex; UN	40	29520.4	6	0.3	2	0.8607
9	Model 8; TOEPH	20	30230.9	8	710.5	20	< 0.0001
10	Model 8; ARH(1)	25	29598.9	8	78.5	15	< 0.0001

M. No.: Number of the model tested; Par.: Number of parameters; -2ℓ: -2 residual log likelihood; Ref.: Model used as reference in the likelihood ratio test; G^2: Test statistic; df: Degrees of freedom; p: p-value. Covariance structures: UN = unstructured, TOEPH = heterogeneous Toeplitz, ARH(1) = heterogeneous first-order autoregressive.

For the reference group (patients 67 years old or younger, not living in a nursing home with a baseline ABPET$<$2.5 and baseline CDRSB$\leq$10), the baseline BPRS score was 58.15 on average (Table 4). Age group had the largest average effect on the baseline BPRS, with an increase of 15.33 for patients between 68 and 77 years old and of 22.99 for patients 78 or older, compared to the youngest group and with the other factors kept constant. Patients living in a nursing home had on average a baseline BPRS that was 3.29 higher than patients not in a nursing home, when the other factors were constant. The higher baseline ABPET group had a baseline BPRS on average 10.17 higher than the lower ABPET group. BPRS increased over time, with 7.16 each additional year in the low baseline CDRSB group and 6.87 in the high baseline CDRSB group.

Table 4: Parameter estimates for the final multivariate model.

Effect	**Estimate**	**Standard Error**	t **Value**	P**-value**
Intercept	58.15	0.328	177.47	< 0.0001
Age 68–77	15.33	0.421	36.43	< 0.0001
Age $\geq$78	22.99	0.840	27.36	< 0.0001
Nursing home	3.29	0.410	8.01	< 0.0001
ABPET $\geq$2.5	10.17	0.655	15.53	< 0.0001
CDRSB $>$10	-0.71	0.397	1.80	0.0727
Year $\times$ CDRSB $\leq$10	7.16	0.035	203.98	< 0.0001
Year $\times$ CDRSB $>$10	6.87	0.056	122.58	< 0.0001

3.4 Two-stage analysis

The F-meta test for a model extension with a quadratic term resulted in a p-value of 0.3219 which did not show evidence to reject the null hypothesis of implementing a linear model, instead of opting for the alternative hypothesis of implementing a quadratic model. The R^2_{Meta} was 0.9892 and 0.9921 for both the linear and quadratic model, respectively (Figure 7). Although the quadratic model had a slightly higher R^2_{Meta}, this increase was very small and to be expected because of the inclusion of an extra parameter. In this context, such marginal gains do not justify the reduction in degrees of freedom and potential overfitting that would come with the inclusion of a quadratic year term.

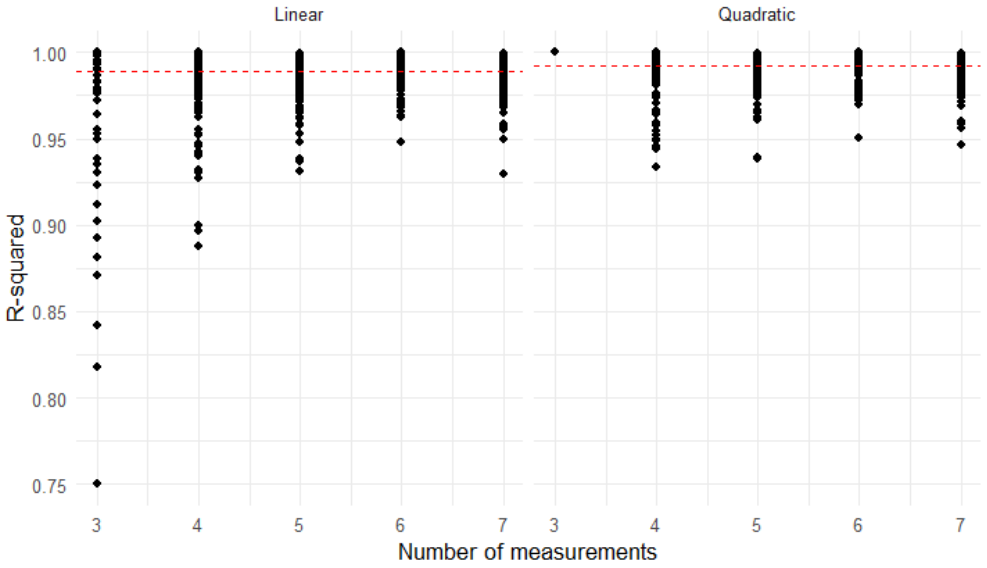


Figure 7: R^2_{Meta} for linear and quadratic stage1 models

Therefore, a separate linear regression of BPRS on time was fitted for each subject in stage 1. This yielded subject-specific baseline BPRS values (intercepts) and yearly rates of change (slopes). The intercepts showed substantial variability across patients, whereas slopes varied only modestly, suggesting that most between-patient heterogeneity was present at baseline rather than in longitudinal trajectories.

In Stage 2, these subject-specific intercepts and slopes were regressed on baseline characteristics (sex, WZC status, age group, baseline CDR-SB group, baseline ABPET group). The model for intercepts had a high R^2 (0.82), indicating that baseline covariates explained most of the heterogeneity in initial BPRS levels. Based on Type III tests p-values suggest that older age groups, residing in a WZC, and higher ABPET levels were associated with markedly higher baseline BPRS, while sex and baseline CDR-SB had no clear effect.

Table 5: Model fit summary for Stage 2 regression of subject-specific intercepts on baseline covariates.

R-Square	Coeff Var	Root MSE	bprs_intercept Mean
0.8171	8.1274	5.9379	73.0604

Table 6: Parameter estimates for Stage 2 regression model explaining subject-specific intercepts.

Parameter	Estimate	Standard Error	t Value	P-value
Intercept	68.43	0.77	88.64	< 0.0001
Female	-0.18	0.37	-0.50	0.62
68–77 years	15.35	0.43	35.91	< 0.001
≥78 years	23.01	0.84	27.18	< 0.001
Nursing home	3.26	0.41	7.91	< 0.001
CDRSB > 10	-0.74	0.40	-1.84	0.06
ABPET ≥ 2.5	10.2090	0.66	-15.45	< 0.001

In contrast, the model for slopes had a very low R^2 (0.017), showing that baseline characteristics explain little about how BPRS changes over time. WZC status (slightly faster worsening, +0.19/year) and baseline CDR-SB (slightly smaller worsening, –0.29/year) had statistically significant small effects on slopes. Based on Type III tests p-values suggest that sex, age group, and baseline ABPET showed no evidence of influencing the rate of change in BPRS. Overall, the two-stage analysis indicates that baseline BPRS differences are strongly related to patient characteristics, while longitudinal BPRS decline was relatively uniform across patients and only weakly associated with baseline variables.

The low R^2 for the stage 2 model for slopes is in line with the results of the multivariate model and the random effects model that showed more variability explained by intercepts than slopes. Moreover, slopes could be noisier when computed via subject-specific ordinary least squares (OLS), because each patient's slope is estimated using only seven points, or less, in the case of drop-out.

Table 7: Model fit summary for Stage 2 regression of subject-specific slopes on baseline covariates.

R-Square	Coeff Var	Root MSE	bprs_slope Mean
0.0172	17.0413	1.2080	7.0891

Table 8: Parameter estimates for Stage 2 regression model explaining subject-specific slopes.

Parameter	Estimate	Standard Error	t Value	P-value
Intercept	7.14	0.15	45.45	< 0.0001
Female	-0.05	0.07	-0.71	0.47
68–77 years	0.006	0.08	0.08	0.93
≥78 years	-0.02	0.172	-0.15	0.88
Nursing home	0.19	0.08	2.27	0.02
CDRSB > 10	-0.29	0.08	-3.58	0.0004
ABPET ≥ 2.5	0.006	0.13	-0.05	0.96

3.5 Analysis with a random-effects model

The preliminary mean structure included as covariates sex, age group, nursing home status, baseline CDRSB, and ABPET. The random-effects structure incorporated a random intercept, a random slope, and a quadratic time effect. Unbiased estimates of the fixed effects were obtainable regardless of the covariance structure. Different covariance structures were evaluated to account for the within-subject correlation over time. Because the exploratory analysis suggested a decreasing correlation over time, models with unstructured (UN), Toeplitz, and ARH(1) covariance structures were compared using restricted maximum likelihood estimation

(REML). Likelihood ratio tests based on REML indicated that the UN model fit significantly better than the Toeplitz model (df = 3, $p < 0.0001$) (Table 9). The ARH(1) did not differ significantly from the UN model (df = 2, p = 0.31). Because the ARH(1) structure achieved a comparable fit with fewer parameters, it was selected as the final covariance structure for the random-effects model. Examination of the variogram suggested the possibility of serial correlation, so the residual covariance structure was further evaluated using alternative functions incorporating measurement error with either a Gaussian or an exponential correlation. Neither of these alternative structures improved model fit, so the residual covariance structure was retained as a simple measurement error model. Additionally, the variogram indicated that these two sources of variability were smaller than the total variability.

Table 9: Overview of likelihood ratio test results (based on REML) for covariance structure selection.

Model	Covariance	Parameter	-2ℓ	df	G^2	p-value
Model 1	UN	7	29588.5			
Model 2	Toep	4	36100.5	3	6512.0	< 0.0001
Model 3	ARH(1)	5	29590.8	2	2.3	0.31

Note: Parameters reflect only covariance (random-effects) and D-matrix rank; fixed-effect parameters are not shown.

The formulation of a plausible random-effects model started with a full model including both linear and quadratic time effects, allowing for random intercepts and slopes as appropriate. To assess whether the quadratic term improved model fit, a likelihood ratio test based on REML compared the full model with a reduced model that excluded the quadratic term. The test indicated that the inclusion of the quadratic term did not improve the model fit, which was consistent with the exploratory data analysis and the F-meta test in the 2 stage model. Therefore, the simpler model with only a linear time effect was retained.

Next, the necessity of including a random slope in addition to the random intercept was evaluated through a likelihood ratio test based on REML, comparing a model with both a random intercept and slope to a model with only a random intercept. The inclusion of a random slope significantly improved the model fit (df = 2, $p < 0.0001$; Table 10). Therefore, the final random-effects model included both a random intercept and a random slope, along with the linear time effect in the fixed part of the model.

Table 10: Overview of likelihood ratio test results (based on REML) for reduction of preliminary random-effects structure.

Model	Random effect	Parameter	-2ℓ	df	G^2	p-value
Model 1	Intercept + slope	6	29590.8			
Model 2	Intercept	4	31485.8	2	1895	< 0.0001

Note: Parameters reflect only covariance (random-effects) and D-matrix rank; fixed-effect parameters are not shown.

After selecting the final covariance structure (ARH(1)) and random-effects structure (random intercept and slope), the next step was to find the most parsimonious mean structure. The full model included all covariates (sex, age group, nursing home status, baseline CDRSB, ABPET, and linear time effect) and likelihood ratio tests were used to compare nested models, sequentially removing non-significant covariates. The results of fitting the full model indicated that the time effects of baseline ABPET, WZS, sex, and age group were not significant. Therefore, these time effects were sequentially removed to simplify the mean structure. Removing the baseline ABPET time effect (Model 1), the WZS time effect (Model 2), the age group time effect (Model 3), and the sex time effect (Model 4) did not lead to a significant loss of fit (all $p > 0.05$), indicating that these time effects were unnecessary and their removal improved model parsimony without compromising fit (Table 11).

Next, the need for including intercepts for the covariates was evaluated. Among the main effects, only the intercepts for sex and baseline CDRSB were non-significant. Because the model already included random slopes for CDRSB, its intercept was retained to preserve interpretability of the slope. Finally, testing the intercept for sex (Model 5) showed that removing this term improved model fit (df = 2, p = 0.86), suggesting that the sex intercept was unnecessary in the presence of other covariates and random effects.

Table 11: Overview of likelihood ratio test results (based on ML) for reduction of preliminary mean structure.

Model	Description	Parameter	-2ℓ	df	G^2	P-value
Full model	All covariates + Time effects	22	29529.6			
Model 1	Full model – baseline ABPET time effect	20	29529.7	2	0.1	0.95
Model 2	Model 1 – WZS time effect	18	29549.4	2	19.7	0.15
Model 3	Model 2 – agegroup time effect	15	29534.0	3	15.5	0.89
Model 4	Model 3 – sex time effect	13	29534.1	2	0.1	0.95
Model 5	Model 4 – sex intercept	11	29534.4	2	0.3	0.86

For patients with a baseline CDRSB value ≤ 10, BPRS changed at a rate of 7.16 units (SE=0.03) per year, while for those with CDRSB > 10, it changed at a slightly lower rate of 6.86 units (SE=0.95) per year; both rates are statistically significant ($p < 0.0001$) (Table 12). Age group also significantly differed in baseline BPRS; patients aged 68–77 had an average baseline BPRS that was 15.31 (SE=0.45) units higher than those aged 67 or younger, and patients older than 77 had an average baseline BPRS that was 22.97 (SE=0.65) units higher, controlling for other covariates. Baseline BPRS scores were significantly higher among nursing home residents, showing an increase of 3.32 units (SE = 0.33) compared to non-residents ($p < 0.0001$). Furthermore, patients with an ABPET score larger than 2.5 at baseline showed a baseline BPRS that was 10.16 (SE=0.35) units higher than those with ABPET < 2.5 ($p < 0.0001$). The change of BPRS over year was significantly different between CDRSB groups. Patients with a CDRSB ≤ 10 had a slightly higher rate of increase (7.16 units/year) compared to those with a CDRSB > 10 (6.86 units/year) ($p < 0.0001$).

Table 12: Parameter estimates for the final random-effects model

Parameter	Estimate effect	Standard Error	t Value	P-value
Intercept	58.15	0.42	136.78	< 0.0001
$68 - 77$ years	15.31	0.45	33.46	< 0.0001
≥ 78 years	22.97	0.65	35.50	< 0.0001
Nursing Home	3.32	0.33	9.93	< 0.0001
CDRSB > 10	-0.70	0.39	-1.79	0.07
ABPET ≥ 2.5	10.16	0.35	28.62	< 0.0001
CDRSB > 10 slope	6.86	0.05	126.62	< 0.0001
CDRSB ≤ 10 slope	7.16	0.03	201.13	< 0.0001

The final covariance structure indicated that patients had highly variable baseline BPRS scores, as demonstrated by a random intercept variance of 33.47 (Table 13). In contrast, there was minimal variation in individual rates of change over time, indicated by a random slope variance of 0.686. Some measurement error within subjects was present, indicated by a residual variance of 2.70; however, this is small compared to the overall between-subject variance. Repeated measurements within the same patient were slightly correlated, with an autoregressive correlation (ρ) of 0.09 (Table 13). This indicated that measurements taken closer together in time tended to be more similar. Measurements farther apart in time were less correlated, and the correlation decays exponentially with the time lag, consistent with a first-order autoregressive structure.

Table 13: Covariance Parameter Estimates

Covariance Parameters	**Estimate**
Random Intercept Variance	33.46
Random Slope Variance	0.68
Autoregressive Correlation (ρ)	0.09
Residual (Within-Subject) Variance	2.70

3.6 Comparison of Random Effects Model and Multivariate Model

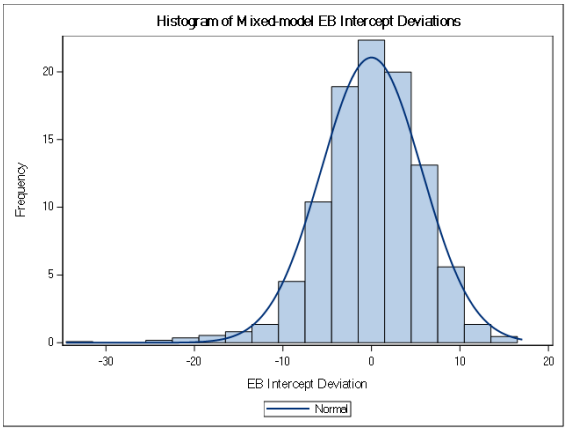
The random-effects model considers both fixed effects (the overall average for everyone) and random effects (individual differences in intercepts and slopes). On the other hand, the multivariate model focuses only on fixed effects (marginal level). These two approaches were compared by looking at the fixed-effect estimates (the average effects). The fixed-effect parameters were very similar in both models (Tables 4 and 12). This shows the multivariate model worked well for this data set and it adjusted the standard errors to account for correlations, however, this model cannot capture individual differences. Moreover, the multivariate model had slightly larger standard errors than the random-effects model. This makes sense because the multivariate approach does not fully account for the variability among individuals, which leads to less precise estimates of the fixed effects.

3.7 Comparison of Subject-Specific Intercepts and Slopes

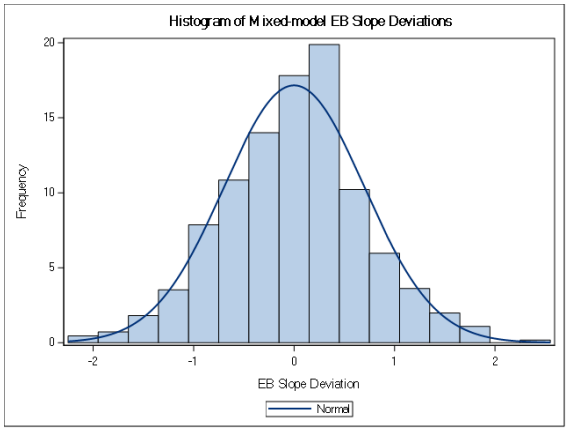
Subject-specific intercepts and slopes were estimated using two approaches: (i) a two-stage method, in which a separate OLS regression of BPRS on time was fitted for each subject, and (ii) a linear mixed model, from which Empirical Bayes (EB) random-intercept and random-slope deviations were obtained. The histograms in Figures 8a-8d summarise the marginal distributions of these effects, while the scatterplots in Figures 9a and 9b display the joint intercept–slope patterns.

The EB intercept deviations (Figure 8a) were tightly concentrated around zero with an approximately normal shape and a relatively narrow spread. The EB slope deviations (Figure 8b) were even more concentrated, with most values close to zero and only limited variation in either direction. The EB intercept–slope scatterplot (Figure 9a) showed a compact cloud of points with the expected shrinkage of EB estimates towards the population mean and suggested a relatively modest degree of true between-subject heterogeneity.

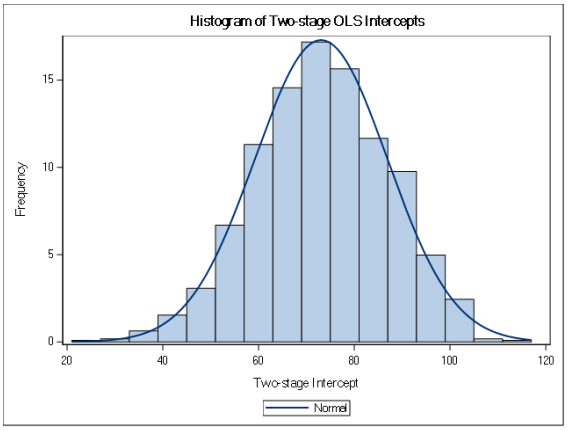
In contrast, the two-stage OLS intercepts (Figure 8c) displayed a much wider distribution, ranging from roughly 20 to 115, with clear extreme cases such as subject 50018. The histogram of two-stage slopes (Figure 8d) also revealed substantially greater spread, with slopes extending from about -2 to 12, again indicating a number of highly atypical subjects. The corresponding intercept–slope scatterplot (Figure 9b) was highly dispersed, with many points far from the main cloud. This reflects the instability of subject-specific OLS fits, particularly for patients with sparse or noisy data, and suggests that the two-stage approach exaggerates the apparent between-subject variability.



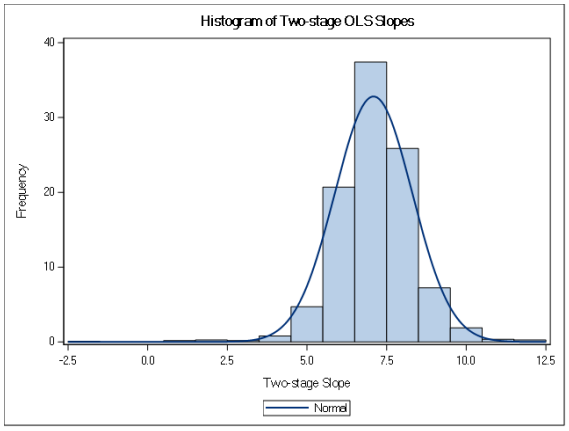
(a) EB intercept deviations



(b) EB slope deviations

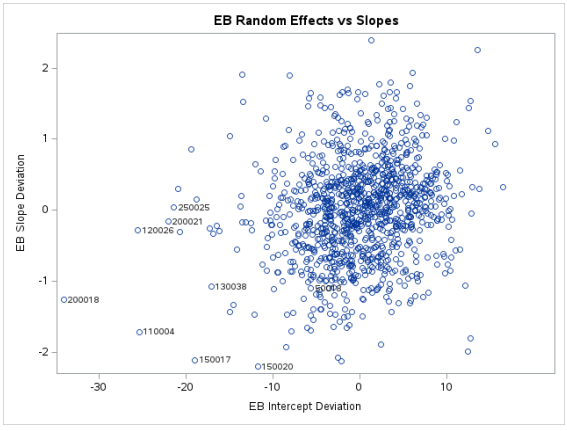


(c) Two-stage OLS intercepts

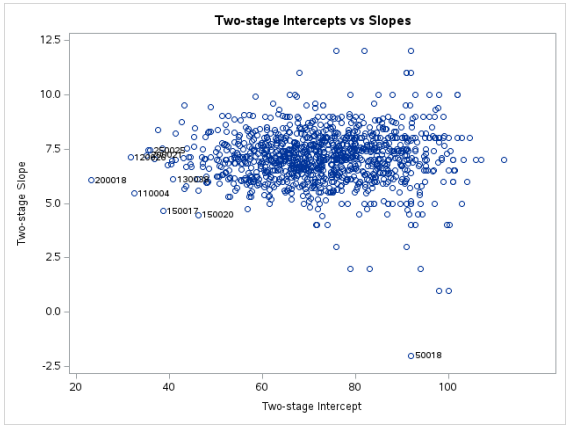


(d) Two-stage OLS slopes

Figure 8: Distributions of subject-specific intercepts and slopes from the mixed-model EB estimates (top row) and the two-stage OLS approach (bottom row).



(a) EB intercept–slope scatterplot



(b) Two-stage intercept–slope scatterplot

Figure 9: Joint distribution of subject-specific intercepts and slopes under the mixed-model EB approach (left) and the two-stage OLS approach (right).

4 Concluding remarks

The analysis of this follow-up study showed that summary statistics analysis were insufficient to give reliable results because they ignored changes within each patient over time. However, the three other analysis methods all point to the same findings about how BPRS scores change in patients with Alzheimer's disease.

The BPRS scores increased linearly over time. This increase was positively associated with baseline CDR-SB measurements: patients with lower baseline CDR-SB values had a faster increase in BPRS during the follow-up period. Other baseline patient characteristics (Age group, Nursing Home, baseline value of ABPET) had only a positive association with the baseline BPRS level and did not influence how BPRS changed over time.

5 References

Wang, Y., Zhan, Y., Yu, E., 2024. Targeted examination of amyloid beta and tau protein accumulation via positron emission tomography for the differential diagnosis of Alzheimer's disease based on the A/T(N) research framework. Clinical Neurology and Neurosurgery 236, 108071.

O'Bryant, S.E., Waring, S.C., Munro Cullum, C., Hall, J., Lacritz, L., Massman, P.J., Lupo, P.J., Reisch, J.S., Doody, R., 2008. Staging Dementia Using Clinical Dementia Rating Scale Sum of Boxes Score. Arch. Neurol. 65:1091-1095.

6 Appendix

```
#----- Exploring the Variance Function -----#
proc loess data=BPRS_long plots=none;
  model BPRS = time;
  output out=BPRS_residuals predicted=meanBPRS residual=residBPRS;
run;

data BPRS_residuals_sq;
  set BPRS_residuals; residBPRS_sq = residBPRS**2;
run;

ods graphics / loessmaxobs=10000;

/* Optional: visualize squared residuals over time */
proc sgplot data=BPRS_residuals_sq;
  scatter x=time y=residBPRS_sq / markerattrs=(symbol=CircleFilled color=blue size=0.8);
  loess x=time y=residBPRS_sq / lineattrs=(thickness=2 color=black);
  xaxis label="Time (years)";
  yaxis label="Squared Residuals";
  title "Squared Residuals of BPRS over Time with LOESS Smoothing";
run;

#----- Semi-variogram -----#
/* Calculation of residuals, linear average trend */
proc glm data=BPRS_long;
model BPRS =time;
output out=out r=residual;
run;

/* Calculation of the variogram */
proc variogram data=out outpair=out;
coordinates xc=time yc=patid;
```

```
compute robust novariogram;
var residual;
run;

data variogram;set out;
if y1=y2;vario=(v1-v2)**2/2; run;
data variance;set out;
if y1<y2; vario=(v1-v2)**2/2; run;

/* Calculation of the total variance (=0.148) */
proc means data=variance mean;
var vario;
run;

/* Loess smoothing of the variogram */
proc loess data=variogram;
ods output scoreresults=out;
model vario=distance;
score data=variogram;
run;

proc sort data=out;by distance;run;


filename fig1 'd:\path\file.eps';
goptions reset=all ftext=swiss device=psepsf gsfname=fig1
gsfmode=replace rotate=landscape;

proc gplot data=out;
plot vario*distance=1 p_vario*distance=2
/ overlay haxis=axis1 vaxis=axis2 vref=295.35 lvref=3;
symbol1 c=red v=dot h=0.2 mode=include;
symbol2 c=black i=join w=2 mode=include;
axis1 label=('Time lag') order=(0 to 7 by 1) minor=none;
axis2 label=(angle=90 'v(u)') order=(0 to 400 by 100) minor=none;
run;quit;

#----- Analysis through the use of summary statistics -----#
/* remove patients with missing measurements */
proc sql;
    create table alzheimer25_complete as
    select *
    from alzheimer25_long
    where patid not in (
        select distinct patid
        from alzheimer25_long
        where bprs is missing
    );
quit;

/* Sort by patient and year */
proc sort data=alzheimer25_complete;
    by patid year;
run;
```

```
/* Calculate AUC, delta BPRS, last BPRS*/
data auc_bprs;
    set alzheimer25_complete;
    by patid;
    retain prev_value auc_sum first_bprs;

    if first.patid then do;
        auc_sum = 0;
        prev_value = bprs;
        first_bprs = bprs;
    end;
    else do;
        /* Trapezoidal AUC ( t =1) */
        auc_sum + (prev_value + bprs) / 2;
        prev_value = bprs;
    end;

    /* Output at last record per patient */
    if last.patid then do;
        last_bprs = bprs;
        delta_bprs = last_bprs - first_bprs;
        output;
    end;

    keep patid auc_sum delta_bprs last_bprs sex agegroup wzc basecdrsbgroup baseabpetgroup
run;

proc glm data=auc_bprs;
    class sex (ref='0')
          agegroup (ref='1')
          wzc (ref='0')
          basecdrsbgroup (ref='0')
          baseabpetgroup (ref='1');
    model auc_sum = sex agegroup wzc basecdrsbgroup baseabpetgroup / solution;
run;
/* The same model for delta_bprs and last_bprs responses */

/* Multivariate model: full model */

PROC MIXED data=datasets.alzheimerlong method=ml;
  class patid sex yearclass agegroup wzc baseabpetgroup basecdrsbgroup;
  model bprs = sex agegroup wzc baseabpetgroup basecdrsbgroup year*sex year*agegroup year*v
  repeated yearclass / type=un subject=patid;
  RUN;

/* Multivariate model: final reduced model */
PROC MIXED data=datasets.alzheimerlong method=ml;
  class patid yearclass wzc agegroup baseabpetgroup basecdrsbgroup /ref=FIRST;
  model bprs = agegroup wzc baseabpetgroup basecdrsbgroup year*basecdrsbgroup / s;
  repeated yearclass / type=un subject=patid;
  RUN;
```

```
#----- Analysis with a random-effects model-----#
proc reg data=alzheimer25_long noprint outest=estimates;
    by patid;
    model bprs = year;
    /* optionally add time  (i.e. year ) if curvature is suspected */
run;

/* Keep only the intercept and slope estimates */
data stage1;
    set estimates(keep=patid intercept year);
    rename intercept = bprs_intercept
           year = bprs_slope;
run;

/* 2. Join Stage 1 & baseline */
/* 2.1 Get one record per subject with baseline covariates */
proc sort data=alzheimer25_long out=baseline nodupkey;
    by patid;
run;

data baseline;
    set baseline;
    keep patid sex agegroup wzc basecdrsbgroup baseabpetgroup;
run;

/* 2.2 Merge Stage1 slopes/intercepts with baseline info */
proc sort data=stage1;
    by patid;
run;

data stage2;
    merge stage1(in=a) baseline(in=b);
    by patid;
    if a and b; /* keep only subjects present in both */
run;

/* 3. Stage 2: Model how intercept/slope depend on baseline covariates */
/* Now you analyze how these subject-level estimates depend on the baseline covariates: */
proc glm data=stage2;
    class
        sex             (ref='0')
        agegroup        (ref='1')
        wzc             (ref='0')
        basecdrsbgroup  (ref='0')
        baseabpetgroup  (ref='1')
        / param=ref;

    model bprs_intercept =
        sex agegroup wzc basecdrsbgroup baseabpetgroup
        / solution;
run;
```

```
proc glm data=stage2;
    class
        sex              (ref='0')
        agegroup         (ref='1')
        wzc              (ref='0')
        basecdrsbgroup   (ref='0')
        baseabpetgroup   (ref='1')
        / param=ref;

    model bprs_slope =
        sex agegroup wzc basecdrsbgroup baseabpetgroup
        / solution;
run;

/Plots: Stage 1/
/Histograms/Boxplots of intercept and slope estimates/
proc sgplot data=stage1;
    title "Histogram of BPRS Intercept";
    histogram bprs_intercept;
    density bprs_intercept;   /* optional smooth curve */
run;

proc sgplot data=stage1;
    title "Boxplot of BPRS Intercept";
    vbox bprs_intercept;
run;

proc sgplot data=stage1;
    title "Histogram of BPRS Slope";
    histogram bprs_slope;
    density bprs_slope;
run;

proc sgplot data=stage1;
    title "Boxplot of BPRS Slope";
    vbox bprs_slope;
run;

/Scatterplot of intercept and slope estimates/
proc sgplot data=stage1;
    scatter x=bprs_intercept y=bprs_slope;
run;

/Plot individual subject regression lines/
data fitted_lines;
    merge mydata(in=a) stage1(in=b);
    by patid;
    if a and b;
    bprs_pred = bprs_intercept + bprs_slope * year;
run;

proc sgplot data=fitted_lines;
    title "Individual Trajectories (Subject-Level Regression Lines)";
```

```
    series x=year y=bprs_pred / group=patid transparency=0.6;
run;

/Plots Stage 2/
/Boxplots of intercept estimate by sex & wzc/
proc sgplot data=stage2;
    vbox bprs_intercept / category=sex;
run;

proc sgplot data=stage2;
    vbox bprs_intercept / category=wzc;
run;

/*full model for reduction of mean*/
proc mixed data=alzheimer25_long method=ml empirical;
    class patid yearclass sex agegroup wzc basecdrsbgroup baseabpetgroup;
    model bprs = sex agegroup wzc basecdrsbgroup baseabpetgroup
                 sex*year agegroup*year wzc*year basecdrsbgroup*year baseabpetgroup*year
                 / noint solution;
    random intercept year/ type=arh(1) subject=patid;
    repeated yearclass / type=simple subject=patid;
run;

/*remove baseabpetgroup time effect*/
proc mixed data=alzheimer25_long method=ml empirical;
    class patid yearclass sex agegroup wzc basecdrsbgroup baseabpetgroup;
    model bprs = sex agegroup wzc basecdrsbgroup baseabpetgroup
                 sex*year agegroup*year wzc*year basecdrsbgroup*year
                 / noint solution;
    random intercept year/ type=arh(1) subject=patid;
    repeated yearclass / type=simple subject=patid;
run;

proc mixed data=alzheimer25_long method=ml empirical;
    class patid yearclass sex agegroup wzc basecdrsbgroup baseabpetgroup;
    model bprs = sex agegroup wzc basecdrsbgroup baseabpetgroup
                 sex*year agegroup*year  basecdrsbgroup*year
                 / noint solution;
    random intercept year/ type=arh(1) subject=patid;
    repeated yearclass / type=simple subject=patid;
run;

proc mixed data=alzheimer25_long method=ml empirical;
    class patid yearclass sex agegroup wzc basecdrsbgroup baseabpetgroup;
    model bprs = sex agegroup wzc basecdrsbgroup baseabpetgroup
                 sex*year  basecdrsbgroup*year
                 / noint solution;
    random intercept year/ type=arh(1) subject=patid;
    repeated yearclass / type=simple subject=patid;
run;

proc mixed data=alzheimer25_long method=ml empirical;
    class patid yearclass sex agegroup wzc basecdrsbgroup baseabpetgroup;
    model bprs = sex agegroup wzc basecdrsbgroup baseabpetgroup
```

```
                basecdrsbgroup*year
                 / noint solution;
    random intercept year/ type=arh(1) subject=patid;
    repeated yearclass / type=simple subject=patid;
run;

/* final reduced model*/
proc mixed data=alzheimer25_long method=ml empirical;
    class patid yearclass sex (ref='0') agegroup (ref='1') wzc (ref='0') basecdrsbgroup (re
    model bprs = agegroup wzc basecdrsbgroup baseabpetgroup
                basecdrsbgroup*year
                 / noint solution;
    random intercept year/ type=arh(1) subject=patid;
    repeated yearclass / type=simple subject=patid;
run;


/* final reduced model */
proc mixed data=alzheimer25_long method=ml empirical;
    class patid yearclass sex (ref='0') agegroup (ref='1') wzc (ref='0') basecdrsbgroup (re
    model bprs = agegroup wzc basecdrsbgroup baseabpetgroup
                basecdrsbgroup*year
                 / solution;
    random intercept year/ type=arh(1) subject=patid;
    repeated yearclass / type=simple subject=patid;
run;

/*-------------------------------------------------------------*/
/* Final mixed model + save random effects (BLUPs)             */
/*-------------------------------------------------------------*/

proc mixed data=alzheimer25_long method=ml empirical;
    class patid yearclass
          sex (ref='0')
          agegroup (ref='1')
          wzc (ref='0')
          basecdrsbgroup (ref='0')
          baseabpetgroup (ref='1');

    model bprs = agegroup wzc basecdrsbgroup baseabpetgroup
                basecdrsbgroup*year
                / solution;

    /* subject-specific random intercept & slope */
    random intercept year / type=arh(1) subject=patid solution;
    repeated yearclass / type=simple subject=patid;

    /* save random effects (EB estimates) */
    ods output SolutionR = RandEff;
run;

/*-------------------------------------------------------------*/
/* Keep only random intercept and random slope (year)          */
/*-------------------------------------------------------------*/
```

```
proc sort data=RandEff;
    by patid Effect;
run;

data RandEff_IS;
    set RandEff;
    where Effect in ('Intercept','year');
    keep patid Effect Estimate;
run;

/*-------------------------------------------------------------------*/
/* Wide format: one row per subject with RE_Intercept &   */
/* RE_year (subject-specific deviations)                  */
/*-------------------------------------------------------------------*/

proc transpose data=RandEff_IS out=RandEff_wide prefix=RE_;
    by patid;
    id Effect;      /* Intercept, year */
    var Estimate;
run;

/* RandEff_wide now has:
      patid
      RE_Intercept = random intercept for subject i
      RE_year      = random slope (year) for subject i          */

/*-------------------------------------------------------------------*/
/* Merge with two-stage subject-specific estimates        */
/*(stage1: bprs_intercept & bprs_slope from PROC REG)     */
/*-------------------------------------------------------------------*/

proc sort data=stage1;
    by patid;
run;

proc sort data=RandEff_wide;
    by patid;
run;

data Compare_Subject;
    merge stage1(in=a) RandEff_wide(in=b);
    by patid;
    if a and b;
run;

/* Compare_Subject contains:
       patid
       bprs_intercept  = two-stage OLS intercept
       bprs_slope      = two-stage OLS slope
       RE_Intercept    = mixed-model random intercept
       RE_year         = mixed-model random slope

#------ Two-Stage analysis (Fmeta)------#
```

```
library(tidyverse)
library(readr)

df <- read_delim("alzheimerlong.txt")

# Define the parameters based on Slide 153
p <- 2      # Number of parameters in the reduced linear model (intercept, time)
p_star <- 1 # Number of added parameters (i.e. time^2)

# 1. For each patient, calculate the Sum of Squared Errors (SSE) for both models
f_meta_data <- df %>%
  filter(!is.na(bprs)) %>%
  group_by(patid) %>%
  mutate(n_obs = n()) %>%
  # Slide 153 rule: Only include patients where n_i >= p + p*
  # In our case, that's n_i >= 2 + 1, so n_i >= 3
  filter(n_obs >= (p + p_star)) %>%
  # Calculate SSE(R) and SSE(F) for each patient
  summarise(
    # Reduced Model (R): bprs ~ year
    # SSE = sum(residuals(model)^2)
    sse_R = sum(residuals(lm(bprs ~ year))^2),

    # Full Model (F): bprs ~ year + I(year^2)
    sse_F = sum(residuals(lm(bprs ~ year + I(year^2)))^2),

    # Get the number of observations for this patient (n_i)
    n_i = first(n_obs)
  ) %>%
  ungroup()

# 2. Calculate the F-meta statistic (using the formula from Slide 153)

# Get the total sums needed for the formula
sum_sse_R_minus_sse_F <- sum(f_meta_data$sse_R - f_meta_data$sse_F)
sum_sse_F <- sum(f_meta_data$sse_F)
Ng <- nrow(f_meta_data) # Total number of patients (N) in the test
sum_df_denominator <- sum(f_meta_data$n_i - p - p_star)

# F-meta formula>Nominator
f_meta_numerator <- (sum_sse_R_minus_sse_F) / (Ng * p_star)

# F-meta formula>Denominator
f_meta_denominator <- sum_sse_F / sum_df_denominator

# F-meta Statistic
F_meta <- f_meta_numerator / f_meta_denominator

# 3. Calculate the p-value for the F-meta statistic
df1 <- Ng * p_star
df2 <- sum_df_denominator

p_value_f_meta <- pf(F_meta, df1, df2, lower.tail = FALSE)
```